

PM SHRI SCHOOL
Jawahar Navodaya Vidyalaya
Monacherra, Hailakandi

Jawahar Navodaya Vidyalaya



Theme — *Science & Technology for Sustainable Future*

Sub-Theme — *Emerging Technologies*

Name — *Shrayan Das*

Class — *XII - Science*

Subject — *Information Technology (IT)*

Guide's Name — *Mr. Rajiv Anand (PGT CS)*

Session — *2026–2027*

EXTINGO

An Autonomous Sensor-Fused Fire Detection & Suppression Robot

1. Overview

Extingo is an Arduino Uno-based robot that continuously scans its surroundings for fire, and on detection, autonomously alerts the owner, ventilates the area, and suppresses the fire with a pressure-adjusted water spray — all while narrating its status on an OLED screen. The design philosophy is sensor fusion with priority weighting: rather than trusting a single sensor, Extingo cross-checks flame, heat and smoke signals in a deliberate order of confidence.

2. System Architecture

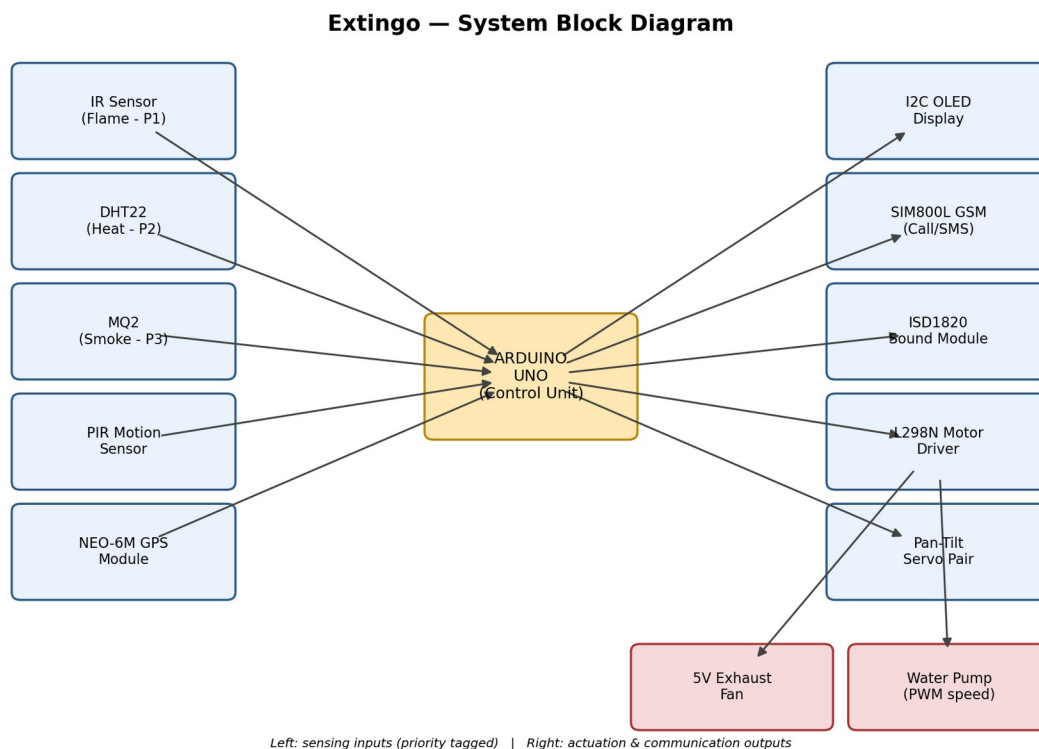


Fig. 1 — All sensors feed decision logic on the Uno; all actuators and comms are driven out from it.

Note on connectivity: "GPS + call/SMS" is implemented as two separate modules sharing the Uno's serial lines (via SoftwareSerial) — a NEO-6M GPS receiver for location, and a SIM800L GSM module for the call and SMS functions — since no single board does both.

3. Initialization & Scanning

On power-up, the I2C OLED immediately prints "Extingo – Ready", confirming a healthy boot before any motion begins. The robot then enters a continuous scan loop driven by two servos in a pan-tilt gimbal:

- Base (pan) servo rotates the whole sensor head in the horizontal plane, sweeping through North, East, South and West headings in fixed angular steps.
- Tilt servo is mounted perpendicular to the pan servo, directly on top of it, and sweeps the elevation angle at each pan stop.
- The IR flame sensor is fixed parallel to the tilt servo's arm, so its line of sight moves exactly with the tilt sweep — giving full azimuth + elevation coverage with only two actuators.

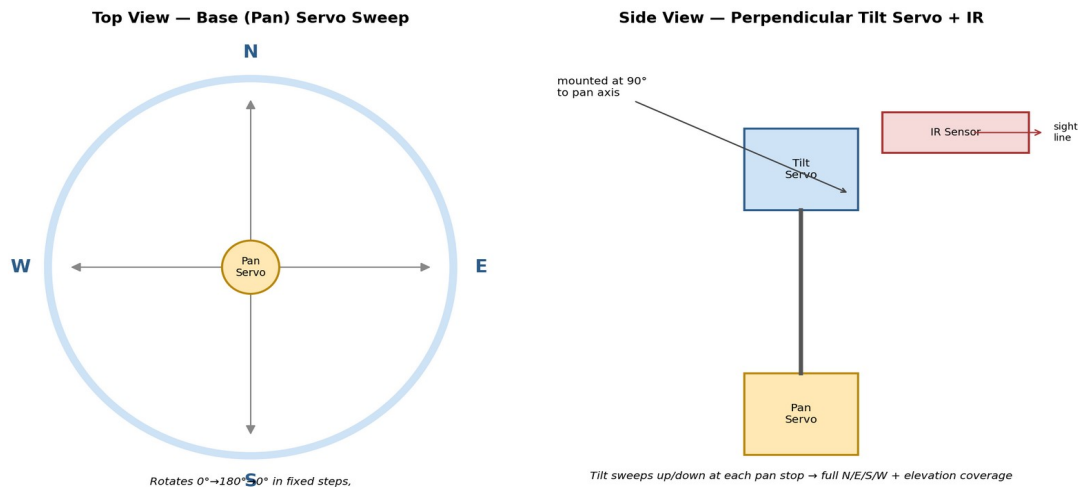


Fig. 2 — Pan covers the four cardinal directions; the perpendicular tilt adds vertical coverage at each stop.

At every stop the Uno reads the IR sensor. If clear, the gimbal advances to the next angle; the loop repeats indefinitely.

4. Fire Detection Logic (Priority Fusion)

Extingo weighs three sensors in a deliberate hierarchy, reflecting how a fire actually develops — flame and heat appear first, smoke often follows:

- Priority 1 — IR Flame Sensor: fastest, most direct evidence of open flame; triggers an immediate hold.
- Priority 2 — DHT22 Heat Sensor: confirms via absolute temperature and, separately, a rate-of-rise check.
- Priority 3 — MQ2 Smoke Sensor: corroborating signal, weighted lowest since smoke can lag or arise from non-fire sources.

A fire is confirmed as soon as the IR sensor fires; DHT22 and MQ2 readings are logged alongside for evidence and are also independently monitored for the rate-of-rise routine described in Section 6.

5. Fire Response Sequence

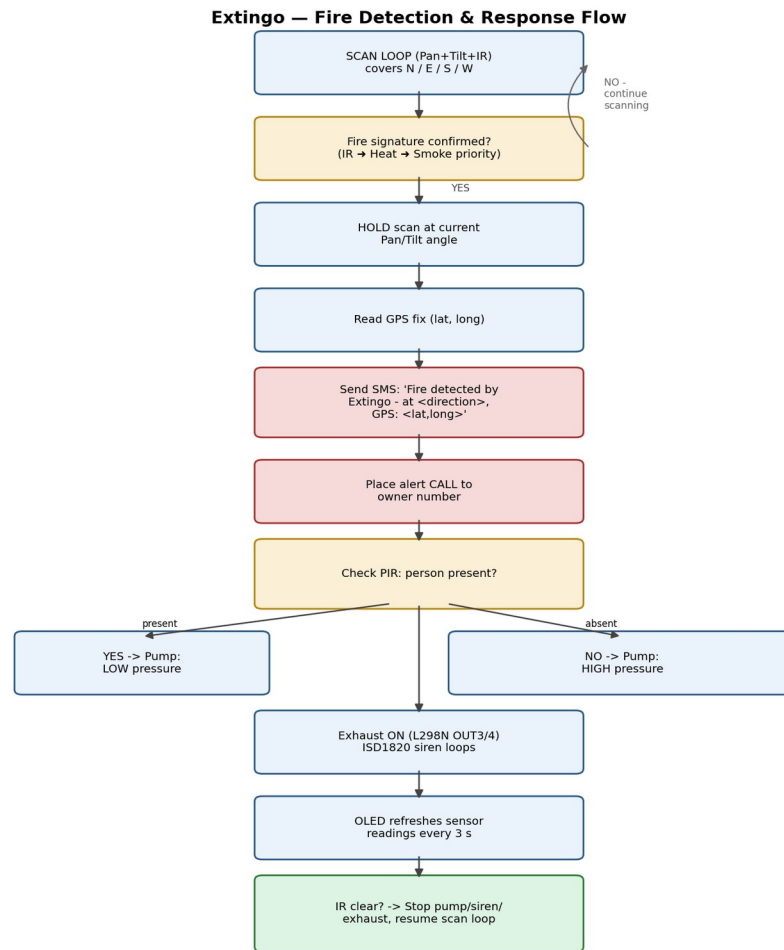


Fig. 3 — End-to-end response, from detection hold through suppression to reset.

Once fire is confirmed, the gimbal holds its current angle (this angle is what gives the direction — N/E/S/W — quoted in the alert). The sequence then runs:

- GPS fix is read and an SMS is sent: "Fire detected by Extingo – at <direction>, GPS: <lat, long>."
- An alert call is placed to the owner's number via the GSM module.
- PIR motion sensor is checked: if a person is present, the water pump runs at LOW pressure (PWM duty reduced) for safety; if the area is empty, it runs at HIGH pressure for faster knockdown.
- The exhaust fan (wired to L298N OUT3/OUT4) switches on to vent smoke and heat.
- The ISD1820 module loops its pre-recorded warning sound for as long as the IR sensor keeps detecting fire.
- The OLED refreshes all live sensor readings every 3 seconds throughout the entire event.

When the IR sensor no longer detects flame, the pump, siren and exhaust are stopped and the robot resumes its scan loop.

6. Rate-of-Rise Heat Algorithm

Independent of flame detection, the Uno samples the DHT22 periodically and computes °C/second between samples. A sudden rise (e.g. beyond a set threshold such as ~2°C/s) is itself treated as an early-warning event, even before the IR sensor confirms open flame:

- SMS sent: "Heat risen by X °C/sec – current temperature Y°C."
- Exhaust fan switches on to start venting immediately.
- When the rate returns to normal and temperature stabilises, exhaust turns off and a confirmation SMS is sent: "All clear – Heat is now low."

This runs as a parallel watchdog alongside the main IR-led detection loop, so Extingo can react to a fast-developing heat event even in the brief window before flame becomes visible to the IR sensor.

7. Component List

Component	Qty	Role in Extingo
Arduino Uno	1	Central controller running the scan, decision and response logic
IR Flame Sensor	1	Priority-1 fire cue; mounted on tilt servo
DHT22	1	Priority-2 cue; also drives rate-of-rise heat algorithm
MQ2 Smoke Sensor	1	Priority-3 corroborating cue
Servo Motor x2	2	Pan (base) + Tilt (perpendicular) scanning gimbal for the IR sensor
I2C OLED Display	1	Status text at boot; live sensor readout every 3 s
NEO-6M GPS Module	1	Supplies lat/long for the SMS alert
SIM800L GSM Module	1	Sends SMS and places the alert call
PIR Motion Sensor	1	Detects occupancy to set pump pressure
Water Pump	1	Suppression; speed varied by PWM (via L298N)
ISD1820 Sound Module	1	Loops a pre-recorded siren while fire persists
5V Exhaust Fan	1	Clears smoke/heat; driven from L298N OUT3/OUT4
L298N Motor Driver	1	Drives exhaust fan and water pump from digital/PWM logic pins

8. Design Rationale (Summary)

- Two-servo gimbal + one IR sensor gives 360° azimuth and elevation coverage at low cost and component count.
- IR > Heat > Smoke priority mirrors real fire progression, reducing false positives from smoke alone.
- PIR-gated pump pressure balances fast suppression against the safety of anyone standing nearby.
- Rate-of-rise DHT22 watchdog catches fast-building heat before flame is visually confirmed.
- GPS-tagged SMS + call + continuous OLED telemetry keep the owner informed in real time, not just at the moment of detection.

thank you